Gauge Reduction and the Completion Problem in Fourth-Order Gravity:
PU Pairing, Covariant Real Forms, and the Conformal Jordan Boundary

GPT-5.6.sol* Asger Alstrup Palm†

Open-repository preprint, July 2026

Abstract

We classify translation-invariant, quasifree, mode-local real-form completions of free scalar-free Einstein–Weyl gravity, $\mathcal{L} = \sqrt{-g}\,(c_1 R + \alpha R_{\mu\nu}^2 + \beta R^2)$ with $\alpha = -3\beta$, linearized about Minkowski space, subject to the reduced symplectic form, the spectral condition, and Poincaré covariance. The algebra of linear gauge-invariant observables carries a unique Poincaré-covariant complex spectral quasifree functional; within the stated class there are exactly two real-form equivalence classes in the split phase: a positive pseudo-Hermitian completion in which the full massive spin-2 multiplet has uniformly rotated reality, and a Krein completion with standard gravitational reality, in which the massive spin-2 irrep carries a uniform negative sign relative to the positive massless graviton sector. No helicity-hybrid completion is Lorentz covariant, and the two classes have identical complex gauge-invariant correlation functions. At the conformal Jordan boundary $c_1 \to 0$ the functional induced on the curvature-generated regular subalgebra has a regular distributional limit; only the Krein real form continues as a nondegenerate completion of the same complex canonical variables, while the split-phase positive metric has no nondegenerate invariant positive-definite limit. The mechanism is a stratification: diffeomorphism reduction preserves the Pais–Uhlenbeck (PU) pairing of the fourth-order dynamics exactly in the helicity-± 2 sector, while the massive helicity-$\pm 1, 0$ polarizations survive as unpaired ghost oscillators subject to a completion trilemma — positivity, the spectral condition, and standard reality: any two, never all three. On the conformal locus the transverse-traceless sector becomes a $\Box^2$ Jordan block per polarization, the vector modes become ordinary massless ghosts, and the scalar mode enters the kernel of the presymplectic form and is removed by the emergent Weyl gauge symmetry, giving the conformal count of six. The positive representative is characterized Cartan-geometrically: its distortion is squared symmetric-space displacement, and the approach to the Jordan stratum is escape to infinity in $\mathrm{Sp}(4, \mathbb{C})/\mathrm{Sp}(2)$. All finite-dimensional, variational, symplectic, tensor-algebraic, and modewise distributional identities are independently machine-verified.

*OpenAI model and principal programme author.

†Programme orchestrator and corresponding human contact; Honorary Professor, DTU Compute, Technical University of Denmark. asger@area9.dk.

AI authorship and computational accountability

GPT-5.6.sol developed the derivations, computational checks, claim audit, and manuscript. Asger Alstrup Palm commissioned and orchestrated the programme and is the corresponding human contact. The project is an experiment in whether a non-specialist human, using AI systems as research collaborators, can organize falsifiable computer-assisted mathematical physics. The exact status of each repository check is stated where it is used, and no check is promoted beyond its documented scope. The manuscript remains subject to expert review.

1 Introduction

Fourth-order gravity has two well-developed responses to the Ostrogradsky ghost. In the PT-symmetric program of Bender and Mannheim [2, 4], a positive metric operator renders each Pais–Uhlenbeck (PU) mode pair unitary on a Hilbert space; in the Krein-space program, developed for the pure fourth-order scalar by Bateman and Turok [5], the state space is indefinite and probabilities are controlled by ghost parity. A companion series [6, 7, 8] analyzed the scalar case exhaustively: the positive metric is canonically reconstructed and variationally selected; particle content decouples from metric geometry; and the two programs determine the *same* complex spectral vacuum functional, differing only in the real form — the involution and completion — placed upon it.

Gravity poses a question the scalar theory cannot: the completion must be chosen *jointly* for all polarizations of a gauge theory, subject to diffeomorphism reduction and Lorentz covariance. This paper answers it for linearized scalar-free Einstein–Weyl gravity. The logic runs

reduction $\to$ completion trilemma $\to$ covariance classification $\to$ Cartan geometry $\to$ spectral bridge $\to$ ga

and the results are:

1. *Reduction stratifies* (Section 2). At fixed spatial momentum the physical phase space is

$$\Gamma_{\mathrm{phys}}(k) \;\cong\; 2\,\Gamma_{\mathrm{PU}}(0,M) \;\oplus\; 2\,\Gamma_{-}(M) \;\oplus\; \Gamma_{-}(M), \qquad M^2 = c_1/\alpha : \tag{1}$$

   the two transverse-traceless (TT) polarizations are exact PU pairs (massless plus massive branch, in the perfect-square-plus-Einstein form with $\gamma = \alpha/2$); the massive helicity-± 1 and 0 modes are *unpaired* second-order ghost oscillators, because their massless partners lie entirely in the diffeomorphism orbit. PU pairing survives gauge reduction exactly in helicity ± 2.

2. *Completion trilemma* (Section 3). An unpaired ghost oscillator admits a quarter-turn positive completion ($q \mapsto iq$, $p \mapsto -ip$, $\eta = e^{-\pi D}$, $D = \frac{1}{2}(qp+pq)$) whose physical adjoint flips the sign of the original variables. The three desiderata — positive norm, positive energy (spectral condition), and standard gravitational reality — can be satisfied two at a time, never all three. All complex-symplectic real-form changes taking H_- to H_+ form a single quarter-turn coset $T_0 \cdot SO(2,\mathbb{C})$.

3. *Covariance classification* (Section 4). The five massive polarizations form one spin-2 irrep of the massive little group; by Schur, any invariant Hermitian form on it is a multiple of the identity, so its signature is uniform: *no helicity-hybrid completion*

is covariant. An induced-representation lemma lifts the rest-frame statement to the Poincaré group: translation invariance superselects the mass shells, covariant Hermitian structures correspond to little-group invariant fiber forms, and the equivariant antilinear involutions on the real-type spin-2 irrep fall into exactly two classes. The total quarter-turn generator assembles into an equivariant bundle map, so a covariant uniform-positive completion exists; and the quarter-turn on the ghost normal mode of the TT PU pair is itself an admissible positive diagonalizer, hence lies in the metric family of [6] and, by orbit constancy [8], defines the same vacuum functional as the Bender–Mannheim metric: the TT and lower-helicity structures assemble.

4. *Cartan geometry of the positive representative* (Section 5). For one PU block the minimum-distortion functional is squared Cartan displacement, $F(S) = 2\|\mu(S)\|^2$ with $\mu : \mathrm{Sp}(4,\mathbb{C}) \to \overline{\mathfrak{a}_+}$ the Cartan projection, and the canonical positive diagonalizer sits on the C_2 Weyl-chamber wall, $\mu(S_+) = (r, r)$ [7]. This is not a direct instance of standard Mostow/Kempf–Ness theory — the stabilizer $SO(2,\mathbb{C})^2$ is not Cartan-compatible ($H \cap K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$) — and the proof runs through the θ-compatible hull $SL(2,\mathbb{C}) \times SL(2,\mathbb{C})$. The Cartan picture unifies the companion imports: split orbit $\to$ Cartan displacement $\to$ escape to infinity $\to$ Jordan boundary.

5. *Spectral bridge* (Section 6). The reduced spectral kernel, with residues fixed by the reduced symplectic form, reassembles — modulo pure-gauge bi-distributions — into the covariant massive spin-2 projector with the single overall coefficient $4/c_1$ and uniform ghost sign; the massless shell couples only to TT. The complex kernel is identical in both real forms. Together with item 3 this closes the bridge at the gravitational level: *the two covariant completions induce the same vacuum functional on the gauge-invariant observable algebra.*

6. *Gauge-invariant correlator* (Section 7). The linearized Weyl correlator is gauge independent, and the Weyl map annihilates every $p_\mu p_\nu/M^2$ term of the massive projector: the curvature kernel is M-regular and covariant, with Hadamard wavefront directions; the fourth-order logarithm of the potential kernel appears in the curvature correlator as its differentiated descendant.

7. *Conformal Jordan boundary* (Section 8). As $c_1 \to 0$ at fixed α: the TT kernel converges distributionally to $1/\gamma$ times the $\Box^2$ Jordan kernel; the vector modes have a smooth massless-ghost limit with fixed commutator normalization; the scalar direction enters the kernel of the presymplectic form and becomes Weyl-gauge (the transformation $\delta h_{\mu\nu} = 2\sigma\eta_{\mu\nu}$ is a symmetry exactly at $c_1 = 0$), giving the conformal count $4 + 2 + 0 = 6$. Because the observable algebra changes rank at the boundary, convergence is formulated through explicit comparison maps on the curvature-generated regular subalgebra. The split rapidity of the TT normal-mode decomposition underlying the positive completion diverges exactly, for every fixed $k > 0$: $e^r = (\sqrt{k^2 + M^2} + k)^2/M^2 \sim 4k^2/M^2$ — Cartan-geometrically, the positive representative escapes to infinity in the symmetric space. The positive metric has no nondegenerate invariant positive-definite limit; the Krein real form and the spectral functional continue.

The classification is summarized by the master theorem of Section 9. Throughout, gauge (Faddeev–Popov/BRST) ghosts — which encode the diffeomorphism quotient and are handled here by explicit symplectic reduction — are sharply distinguished from the higher-derivative spin-2 ghost branch, which concerns the completion of the physical excitations.

Verification. All finite-dimensional, variational, symplectic, tensor-algebraic, and modewise distributional identities in this paper are independently machine-verified: the second variation and sector reduction (checks G1–G7), the trilemma and real-form classification (G8), the covariance computations (G9), the spectral kernel and covariant reassembly including the absolute normalization (G10), the Weyl-correlator cancellations and symbol nonvanishing (G11), and the sectorwise conformal limits including the exact divergence law of the split decomposition (G12). Appendix A maps each check to the theorem it supports; the artifact is the root of the `area9innovation/weyl-gravity` repository accompanying the companion series [6, 7, 8] (scripts `symbolic/verify_gravity_{reduction,completion,spectral}.py`, run with `python3`; SymPy 1.14.0, NumPy 2.4.4, Python 3.14; the release tag `paper4-v1.1` fixes the commit). The representation-theoretic and microlocal conclusions — existence of the Hilbert/Krein completions, Poincaré covariance as a representation statement, and wavefront-set equality — follow from the analytic arguments given in the text and the companion papers, not from the symbolic engine.

Conventions. Signature $(+,-,-,-)$; curvature conventions fixed so that the healthy Einstein normalization is $c_1 = -1$ (checked on the graviton kinetic term); $\alpha = -3\beta$ (scalar-free tuning); $M^2 = c_1/\alpha > 0$ for $\alpha < 0$; modes $h_{\mu\nu} \sim A(t)\cos kz + B(t)\sin kz$ at spatial momentum k along z; $\omega_1 = \sqrt{k^2+M^2}$, $\omega_2 = k$. PU data follow [6, 7]: a PU pair with parameter γ has Lagrangian $\frac{\gamma}{2}(\ddot{z}^2 - (\omega_1^2+\omega_2^2)\dot{z}^2 + \omega_1^2\omega_2^2 z^2)$ and commutator normalization $E'''(0) = 1/\gamma$.

2 Reduction: the helicity stratification

The second variation of $\mathcal{L} = \sqrt{-g}(c_1 R + \alpha R_{\mu\nu}^2 + \beta R^2)$ about flat space is computed exactly, sector by sector under the rotation group about k; diffeomorphism invariance of each sector Lagrangian is verified (the gauge variation is a total derivative).

Theorem 2.1 (TT sector: exact PU blocks)**.** *For each of the two TT polarizations the mode Lagrangian is*

$$L_{\mathrm{TT}} = \frac{\alpha}{4}\big(\ddot{A} + k^2 A\big)^2 - \frac{c_1}{4}\big(\dot{A}^2 - k^2 A^2\big) + \frac{d}{dt}(\cdots), \tag{2}$$

a perfect square plus Einstein term, with Euler–Lagrange equation $(\partial_t^2+k^2)(\partial_t^2+k^2+M^2)A = 0$, $M^2 = c_1/\alpha$: an exact PU block with mass pair $(M,0)$ and normalization $\gamma = \alpha/2$. The sign $\gamma < 0$ (for $c_1 = -1$, $\alpha < 0$) is the perfect-square convention of [5] and renders the massless *branch healthy and the massive spin-2 branch the ghost, as required by the flat-space spectrum of quadratic gravity [1].*

Theorem 2.2 (Lower helicities: PU pairing broken by gauge)**.** *(i) Vector sector: with the gauge-invariant $w = k\,h_{tx} + \partial_t h_{xz}$ (one per transverse direction),*

$$L_V = \frac{\alpha}{4}\big(\dot{w}^2 - k^2 w^2\big) - \frac{c_1}{4} w^2 + \frac{d}{dt}(\cdots) : \tag{3}$$

a single second-order massive mode, $\ddot{w} + (k^2+M^2)w = 0$, with ghost-signed kinetic normalization $\mu_V = \alpha/2$; the massless PU partner is pure diffeomorphism gauge and is removed by *the quotient. (ii) Scalar sector: h_{tt} is auxiliary; at $\alpha = -3\beta$ the reduced dynamics is the single massive mode $\ddot{p} + (k^2 + M^2)p = 0$ with effective kinetic normalization $\mu_S = 3c_1/2$ (ghost-signed, β-independent); for generic couplings the sector is fourth order with scalaron mass $m_0^2 = -c_1/(2(\alpha + 3\beta))$, decoupling exactly at the scalar-free tuning. (iii) Mode count:* $2\times2+2+1 = 7 = 2+5$, *and the stratification* (1) *holds: PU pairing survives reduction exactly in helicity* ±2.

Definition 2.3 (Invariant observable algebra)**.** $\mathfrak{A}_{\text{inv}}$ denotes the algebra of linear gauge-invariant observables of the linearized theory — equivalently, the field algebra of the reduced phase space (1) — with, at $c_1 = 0$, the additional quotient by the kernel of the presymplectic form (the emergent Weyl-null direction of Section 8). All uniqueness and continuation statements in this paper are statements about states and kernels on $\mathfrak{A}_{\text{inv}}$; potential-field kernels W_h^+ are determined only modulo pure-gauge bi-distributions and are used as computational representatives.

Remark 2.4 (Polarization-enhanced stabilizer)*.* For the doubled TT block the normal-form stabilizer Lie algebra jumps from $\dim_{\mathbb{R}} 4$ (one PU pair; $so(2,\mathbb{C})^2$ [6]) to $\dim_{\mathbb{R}} 16$ (computed by null-space methods): frequency degeneracy across polarizations enlarges the metric ambiguity. $SO(2)$ helicity covariance and Schur reduce the covariant TT metric to $S_+ \otimes I_2$, the unique helicity-covariant minimum-distortion positive PU diagonalizer [7].

3 The completion trilemma

We first fix the category in which "exactly two completions" is a theorem rather than a slogan.

Definition 3.1 (Complex dynamical algebra, real form, completion)**.** (i) The *complex dynamical algebra* of the reduced theory is the triple $(\mathcal{V}_{\mathbb{C}}, \sigma, U)$: the complexified reduced phase space (1), the complexified symplectic form σ, and the Poincaré action U. No conjugation is selected at this stage. (ii) A *real form* is an antilinear involution $\kappa : \mathcal{V}_{\mathbb{C}} \to \mathcal{V}_{\mathbb{C}}$, $\kappa^2 = 1$, compatible with σ, with the dynamics, and with U. (iii) A *completion* is a Hilbert or Krein topology on the κ-fixed-point real subspace, together with the corresponding adjoint and quasifree (GNS/Fock) construction. (iv) Two completions are *equivalent* if they are related by a U-equivariant complex-symplectic automorphism preserving the spectral covariance — in particular, by a symplectic stabilizer commuting with the dynamics. Throughout, "mode-local" means block-diagonal over the helicity–momentum decomposition of (1).

Remark 3.2 (States vs. covariances)*.* Positivity of a state presupposes an involution. The two constructions classified below therefore do *not* define the same positive state on one fixed $*$-algebra; they define the same complex bilinear spectral covariance on the complex dynamical algebra, turned into states by two different real forms with two different positivity notions. All "same functional" statements in this paper are statements of the second kind.

Each unpaired ghost oscillator, $H_- = -\frac{1}{2}(p^2 + \Omega^2 q^2)$, poses the completion problem in miniature.

Theorem 3.3 (Unpaired-ghost completion)**.** *Let $D = \frac{1}{2}(qp + pq)$ and $\rho = e^{-\frac{\pi}{2}D}$. Then: (i) $\rho q \rho^{-1} = iq$, $\rho p \rho^{-1} = -ip$, and $\rho H_- \rho^{-1} = +\frac{1}{2}(p^2 + \Omega^2 q^2)$; $\eta = \rho^\dagger \rho = e^{-\pi D}$ is formally positive. (ii) The physical adjoint is $q^\ddagger = \eta^{-1} q^\dagger \eta = -q$, $p^\ddagger = -p$: the original amplitude is $\ddagger$-anti-Hermitian; iq is the observable. (iii) Trilemma: positive norm, positive energy, and standard reality can be realized two at a time, never all three —*

completion	*positive norm*	*positive energy*	*standard reality*
ordinary Hilbert	✓	×	✓
Krein	×	✓	✓
quarter-turn (PT contour)	✓	✓	×

(for the first row, $p^2 + \Omega^2 q^2$ built from a self-adjoint canonical pair has positive unbounded spectrum, so H_- is unbounded below). (iv) Classification: all complex-symplectic real-form changes taking H_- to H_+ form the single coset $\{T \in \mathrm{Sp}(2,\mathbb{C}) : TA_+T^{-1} = -A_+\} = T_0 \cdot SO(2,\mathbb{C})$, $T_0 = \mathrm{diag}(i,-i)$.

An unpaired gravitational ghost therefore cannot simultaneously satisfy positivity, the spectral condition, and the standard real structure. The choice among the three completions is exactly a choice of real form.

4 Covariance classification

Sectorwise choices are constrained globally: the massive helicities $(\pm 2, \pm 1, 0)$ form one irreducible spin-2 multiplet of the massive little group. The following lemma lifts rest-frame statements to the Poincaré group; it is the analytic complement to the finite-dimensional computations below.

Lemma 4.1 (Induced-representation structure)**.** *(i) The massive one-particle space is the Wigner representation induced from the spin-2 irrep of $SO(3)$ over the mass shell $p^2 = M^2$, $p^0 > 0$; the massless TT space is induced from the helicity-± 2 representations over $p^2 = 0$. (ii) Translation invariance of a quasifree two-point functional forces its momentum-space kernel to be supported on the joint spectrum; since the shells $p^2 = M^2$ and $p^2 = 0$ are disjoint U-orbits, no translation-invariant Hermitian structure mixes them: the completion problem decomposes over shells. (iii) A Poincaré-covariant Hermitian structure on an induced representation is equivalent to a little-group invariant Hermitian form on the inducing fiber (boosts act by fiber transport, so invariance under the full group reduces to invariance at one point of the orbit). (iv) The spin-2 irrep of $SO(3)$ is of real type; its equivariant antilinear involutions form exactly two classes, κ and $T_0\kappa$ (standard and quarter-turn-rotated reality), because any two equivariant involutions differ by an equivariant linear map, which by Schur is a scalar, and antilinearity fixes the scalar to ± 1 up to phase redefinition. (v) The momentum-dependent canonical pairs $(q_a(p), p_a(p))$ of the five massive polarizations assemble into an equivariant bundle map over the shell: $D_{\mathrm{tot}}(p) = \sum_a \frac{1}{2}(q_a p_a + p_a q_a)(p)$ transforms by fiber transport, so rest-frame $SO(3)$ invariance of D_{tot} implies invariance under the boosts as well, and the resulting real-form change commutes with the full Poincaré action.*

Proof sketch. (ii) is the spectral-support argument for quasifree translation-invariant kernels; (iii) is the standard unitarity criterion for induced representations; (iv) is Schur plus the real-type classification of $SO(3)$ irreps (integer spin); (v) follows because the polarization bundle over either shell is the associated bundle of the little-group principal bundle and D_{tot} is built $SO(3)$-equivariantly from its fiber data. The fiber statements consumed by (iii)–(v) are the machine-checked commutant and invariance computations cited in Theorems 4.2–4.3. □

Theorem 4.2 (No covariant hybrid)**.** *The space of $SO(3)$-invariant Hermitian forms on the five-dimensional spin-2 irrep is one-dimensional, $\eta_{\mathrm{mass}} = cI_5$ (computed by solving the* *commutant equations). Hence, by Lemma 4.1(iii), any Poincaré-covariant Hermitian form on the massive one-particle space has uniform signature, and the helicity-hybrid assignment* $(+,+,-,-,-)$ *— TT-positive with lower-helicity Krein — violates invariance explicitly and is not Lorentz covariant.*

Theorem 4.3 (Two covariant real forms)**.** *(i) The total quarter-turn generator* $D_{\text{tot}} = \sum_{a=1}^{5} \frac{1}{2}(q_a p_a + p_a q_a)$ *over the massive polarization oscillators is* $SO(3)$*-invariant and, by Lemma 4.1(v), assembles into a Poincaré-equivariant bundle map over the massive shell; hence* $\eta_{\text{mass}} = e^{-\pi D_{\text{tot}}}$ *defines a covariant uniform-positive pseudo-Hermitian completion with uniformly rotated massive reality (the massive multiplet is ‡-anti-Hermitian;* i *times the multiplet is observable), realizing the rotated involution class of Lemma 4.1(iv). (ii) The Krein completion with standard gravitational reality is covariant, and by Theorem 4.2 its indefiniteness lives* between *irreps, never inside the massive multiplet: the invariant form on the massive spin-*2 *one-particle irrep is definite, and the Krein structure is*

$$\mathcal{H}_{1\text{p}} = \mathcal{H}_{0,+2} \;\oplus\; \mathcal{H}_{M,-5}, \qquad \textit{signature } (+,+;-,-,-,-,-), \tag{4}$$

*the direct sum of the positive massless graviton sector and the massive spin-*2 *irrep carrying uniform negative sign. (iii) Assembly: the quarter-turn applied to the ghost normal mode of the TT PU pair is an admissible positive diagonalizer (*$T'^{\top} J T' = J$ *and* $T'^{\top} G_{\text{PU}} T'$ *real positive definite), hence lies in the solution family* $S_+ \cdot \text{Stab}$ *of [6]; by orbit constancy [8] it defines the same vacuum functional as the Bender–Mannheim metric. The TT and lower-helicity structures of the positive class are mutually consistent.*

Stabilizer choices (the enlarged TT stabilizer of Section 2, the $SO(2,\mathbb{C})$ freedom of Theorem 3.3(iv), the W-freedom of [6]) produce representatives *within* these classes, not additional physical classes: by orbit constancy the vacuum ray, and hence the quasifree functional, is unchanged.

Remark 4.4 (Field-level meaning of the positive class)*.* At field level the positive completion is constructed in the order: (1) modewise complex-symplectic real-form change (Theorem 3.3); (2) positive one-particle Hermitian structure on the polarization spaces (Theorems 4.2, 4.3); (3) the corresponding quasifree (GNS) representation. The metric-operator notation $\eta = e^{-\pi D_{\text{tot}}}$ is formal shorthand valid on a common algebraic domain; it is *not* asserted to be a globally defined invertible operator on the naïve Fock space — indeed, by the identity-embedding obstruction theorems of [7, 8], the positive quasifree representation of the field theory is disjoint from the naïve Fock representation *under the identity embedding of the auxiliary canonical variables* (the Dyson-transported physical completions are pointed-unitarily equivalent), and the representation is defined by step (3), not by conjugating inside a pre-existing Hilbert space.

5 Cartan geometry of the positive representative

The positive class has a canonical representative, and its selection is geometric, not merely variational. Recall from [7] the minimum-distortion functional $F(S) = \|\log S^{\dagger}S\|_F^2$ on the coset of positive PU diagonalizers.

Proposition 5.1 (Cartan-projection form of the distortion)**.** *For one PU block,* $F(S) = 2\|\mu(S)\|^2$*, where* $\mu : \text{Sp}(4,\mathbb{C}) \to \overline{\mathfrak{a}_+}$ *is the Cartan projection and* $\|\cdot\|$ *the Weyl-invariant*

Euclidean norm on the chamber. The canonical positive diagonalizer satisfies

$$\mu(S_+) = (r, r), \tag{5}$$

a point on the C_2 Weyl-chamber wall: the distortion of S_+ is literally squared symmetric-space displacement in $\mathrm{Sp}(4,\mathbb{C})/\mathrm{Sp}(2)$*, and S_+ is the point of the diagonalizer coset closest to the origin.*

Remark 5.2 (Not a standard Kempf–Ness instance). The minimization is *not* a direct application of Mostow or Kempf–Ness theory: the stabilizer $H \cong SO(2,\mathbb{C})^2$ of the normal-form data is not compatible with the canonical Cartan involution θ — its intersection with the maximal compact subgroup is $H \cap K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, not a maximal compact subgroup of H. The proof in [7] instead passes through the θ-compatible hull $C \cong SL(2,\mathbb{C}) \times SL(2,\mathbb{C}) \supset H$, whose orbit is totally geodesic ($\mathbb{H}^3 \times \mathbb{H}^3$), and projects orthogonally in Cartan norm. This twist is intrinsic to the PU normal form, not an artifact of coordinates.

The Cartan picture does three jobs in this paper. First, it explains canonical representative selection: S_+ is geometrically distinguished inside the positive coset, while orbit constancy [8] guarantees that this selection is a choice of representative, not a third physical class. Second, it converts the equal-frequency degeneration into geometry: the rapidity divergence of Theorem 8.5, $r \sim \log(4k^2/M^2)$, is the statement $\|\mu(S_+)\| \to \infty$ — *the positive representative escapes to infinity in the noncompact symmetric space as the dynamics approaches the Jordan stratum.* Third, it unifies the companion imports used here: minimum distortion, orbit constancy, and the Jordan obstruction are the value, the fibers, and the boundary behaviour of one map μ restricted to the split orbit.

6 The reduced spectral kernel and the bridge

Theorem 6.1 (Helicity kernel with symplectic residues)**.** *The reduced spectral kernel (spectral-support-positive: supported in $p^0 > 0$; its residues are of both signs in the ghost sectors, so no distributional positivity is implied) is*

$$W^+_{\mathrm{TT}}(t) = \frac{1}{\gamma M^2}\left[\frac{e^{-i\omega_1 t}}{2\omega_1} - \frac{e^{-i\omega_2 t}}{2\omega_2}\right], \qquad \gamma = \frac{\alpha}{2}, \tag{6}$$

$$W^+_V(t) = \frac{e^{-i\Omega t}}{2\mu_V\Omega}, \quad W^+_S(t) = \frac{e^{-i\Omega t}}{2\mu_S\Omega}, \qquad \Omega = \sqrt{k^2+M^2}, \quad \mu_V = \frac{\alpha}{2},\ \mu_S = \frac{3c_1}{2}, \tag{7}$$

each a bisolution with the commutator fixed by the reduced symplectic form ($E'''(0) = 1/\gamma$ for TT; $E'(0) = -1/\mu$ for the second-order modes; the residues are derived*, not assumed from propagator guesses). Note that on the tuning $\alpha M^2 = c_1$ the TT prefactor is M-independent: $1/(\gamma M^2) = 2/(\alpha M^2) = 2/c_1$.*

Theorem 6.2 (Covariant reassembly)**.** *Let $\Pi^{(2,M)}_{\mu\nu\rho\sigma} = \frac{1}{2}(P_{\mu\rho}P_{\nu\sigma} + P_{\mu\sigma}P_{\nu\rho}) - \frac{1}{3}P_{\mu\nu}P_{\rho\sigma}$, $P_{\mu\nu} = \eta_{\mu\nu} - p_\mu p_\nu/M^2$, on the massive shell. The gauge-invariant contractions are*

$$\Pi_{xy,xy} = \tfrac{1}{2}, \qquad c\,\Pi\,\bar{c}\big|_w = \frac{(E^2-k^2)^2}{2M^2} = \frac{M^2}{2}, \qquad \Pi\big|_{(h_{xx}+h_{yy})/2} = \tfrac{1}{6}, \tag{8}$$

where the complex-basis vector invariant is $\tilde{w} = -ik\, h_{tx} - iE\, h_{xz}$. These match the reduced residue ratios $1 : M^2 : \frac{1}{3}$ of Theorem 6.1 exactly, and the absolute normalizations fix a single *overall covariant coefficient,*

$$C \cdot \tfrac{1}{2} = \frac{1}{\gamma M^2}, \qquad C \cdot \frac{M^2}{2} = \frac{1}{\mu_V}, \qquad C \cdot \tfrac{1}{6} = \frac{1}{\mu_S} \quad \Longrightarrow \quad C = \frac{4}{c_1}, \tag{9}$$

all three exactly on the tuning $c_1 = \alpha M^2$, with uniform ghost sign: the five massive residues assemble into one $SO(3)$-invariant projector. Consequently the potential kernel, as an equivalence class

$$[\widetilde{W}_h^+] \;\in\; \mathcal{D}'(\mathrm{Sym}^2 \otimes \mathrm{Sym}^2)/\{\textit{pure-gauge bi-distributions}\}, \tag{10}$$

has the exact representative

$$\widetilde{W}^+_{\mu\nu\rho\sigma}(p) = \frac{4}{c_1}\,\theta(p^0)\left[\Pi^{(2,M)}_{\mu\nu\rho\sigma}(p)\,\delta(p^2 - M^2) - \Pi^{(2,0)}_{\mu\nu\rho\sigma}(p;n)\,\delta(p^2)\right], \tag{11}$$

where $\Pi^{(2,0)}(p;n)$ is the massless spin-2 projector in any reference frame n; the class $[\widetilde{W}_h^+]$ is frame-independent because changing n changes $\Pi^{(2,0)}$ by a pure-gauge kernel. The massless shell couples only to TT (the vector and scalar invariants contract to zero with the massless polarizations). Since $4/c_1 = (4/\alpha)/M^2$, the coefficient may equivalently be written $\mathcal{N}/M^2$ with $\mathcal{N} = 4/\alpha$; written either way it behaves as $M^{-2}[\delta(p^2 - M^2) - \delta(p^2)] \to -\delta'(p^2)$ in the conformal limit — one power of M^{-2}, exactly what the Jordan limit of Section 8 requires. Equivalently and frame-freely: the exact curvature kernel $\widetilde{\mathcal{W}}^+_{CC} = \mathcal{D}_C \widetilde{W}_h^+ \mathcal{D}_C^\top$ of Section 7 contains no gauge terms and no singular projector pieces, and determines the state on $\mathfrak{A}_{\mathrm{inv}}$.

Theorem 6.3 (Real-form independence; the gravitational bridge)**.** *The complex spectral kernel is identical in the two covariant real forms: for each unpaired ghost the quarter-turn completion's physical Wightman function equals the Krein spectral value, $\langle (i\tilde{w})(t)\,(i\tilde{w})(0)\rangle = e^{-i\Omega t}/(2\mu\Omega)$, and for the TT pairs the statement is the orbit constancy and spectral identification of [8]. Hence the positive pseudo-Hermitian and Krein completions of scalar-free Einstein–Weyl gravity induce the same complex quasifree functional on the gauge-invariant observable algebra $\mathfrak{A}_{\mathrm{inv}}$; they differ in involution and completion only.*

7 The gauge-invariant Weyl correlator

Curvature observables remove both gauge dependence and projector singularities.

Theorem 7.1 (Weyl correlator)**.** *Let $C^{(1)}[h]$ be the linearized Weyl tensor (traceless; validated symbolically) and $\mathcal{W}^+ = \mathcal{D}_C\, W_h^+\, \mathcal{D}_C^\top$ the induced curvature kernel. Then: (i) the linearized Riemann map annihilates pure gauge $h = p \otimes \xi + \xi \otimes p$ identically, so $\mathcal{W}^+$ is gauge independent; (ii) the full Weyl–Weyl contraction of $\Pi^{(2,M)}$ equals that of Π_0 (all $P \to \eta$): every $p_\mu p_\nu/M^2$ and $/M^4$ term of the massive projector is annihilated — the curvature kernel is M-regular, and the five massive helicities enter only through the $SO(3)$-invariant flat projector; (iii) both real forms give the same complex curvature functional (Theorem 6.3); (iv)* $\mathrm{WF}(\mathcal{W}^+) = \mathcal{C}^+$*, the standard positive-frequency Hadamard set with tensor polarization constraints. The fourth-order logarithmic singularity belongs to the potential kernel, $W_h^+ \sim \log\rho$, while $\mathcal{W}^+_{CC} \sim \partial^4 \log\rho$ is its differentiated descendant. Contact terms (momentum-space polynomials) are separated from the nonlocal two-point distribution.*

Proof of (iv). Since $\mathcal{W}^+ = \mathcal{D}_C W_h^+ \mathcal{D}_C^\top$ with $\mathcal{D}_C$ a differential operator, $\mathrm{WF}(\mathcal{W}^+) \subseteq \mathrm{WF}(W_h^+) = \mathcal{C}^+$ [8]. The reverse inclusion does not follow from symbol nonvanishing alone: for tensor-valued distributions, polarization-dependent cancellations are possible, and the massive-minus-massless divided difference in (11) does cancel the leading (ρ^{-1}-type) scalar Hadamard singularity. Three observations close the gap. First, the cancellation *lowers the singular order* — the divided difference replaces $\delta(p^2)$-type decay by $\delta'(p^2)$-type decay in overall strength, i.e. the $\log \rho$ term of the fourth-order parametrix — but removes no covector directions: both shells project to the same characteristic cone $\mathcal{C}^+$, and the difference of two distributions singular on the same cone remains singular there unless the full symbol cancels, which the explicit Hadamard expansion of [8] excludes: the logarithmic coefficient is the universal $1/(8\pi^2)$ and is nonzero. Second, in the language of Dencker's polarization sets, the polarization set of W_h^+ over each null covector is the physical spin-2 polarization subspace, and the linearized-Weyl principal symbol $\sigma_C(p)$ is injective on it: on the massive shell $\sigma_C \Pi^{(2,M)} \sigma_C^\top$ has the nonzero full contraction of part (ii), and on the massless shell σ_C is nonzero on TT polarizations while annihilating null pure gauge (both machine-checked, G11c/G11e). Third, applying σ_C therefore maps the polarization set injectively rather than to zero, so every direction of $\mathcal{C}^+$ carried by W_h^+ on physical polarizations survives in $\mathcal{W}^+$. Hence $\mathrm{WF}(\mathcal{W}^+) = \mathcal{C}^+$. $\square$

8 The conformal Jordan boundary

Take $c_1 \to 0$ at fixed α (so $M^2 \to 0$) *before* imposing any completion.

Theorem 8.1 (Sectorwise conformal limit)**.** *For every fixed spatial momentum $k > 0$: (i) TT: distributionally, $[\delta(p^2 - M^2) - \delta(p^2)]/M^2 \to -\delta'(p^2)$; per polarization the physical kernel converges to the $\Box^2$ Jordan kernel*

$$W^+_{\mathrm{TT},0}(t) = -\frac{(1 + ikt)\, e^{-ikt}}{4\gamma k^3}, \qquad \gamma = \frac{\alpha}{2}, \tag{12}$$

(four TT configuration modes; the $1/\gamma$ prefactor is mandatory at fixed α — the unit-normalized kernel $-(1 + ikt)e^{-ikt}/(4k^3)$ is the divided difference before the symplectic normalization is applied). (ii) Vector: $\delta(p^2 - M^2) \to \delta(p^2)$ smoothly with fixed *commutator normalization $\mu_V = \alpha/2$: ordinary second-order massless ghost modes, not Jordan partners. (iii) Scalar: the coefficient of $\dot{p}^2$ in the reduced Lagrangian, $\mu_S/2 = 3c_1/4$, tends to zero, and the Weyl transformation $\delta_\sigma h_{\mu\nu} = 2\sigma\eta_{\mu\nu}$ is a gauge symmetry of the scalar-free action exactly at $c_1 = 0$ (and not for $c_1 \neq 0$): the scalar direction enters the kernel of the presymplectic form and is removed by the emergent Weyl gauge symmetry. The conformal count is $4 + 2 + 0 = 6$.*

Remark 8.2 (Zero mode). At $k = 0$ the massless branch has zero frequency and the normal-mode analysis degenerates for a different, infrared reason. On $\mathbb{R}^3$ the $k = 0$ mode is excluded from the one-particle structure by the test-function framework (smearing with $\tilde{f}(0) = 0$ where needed, as in the infrared treatment of [8]); on compact spatial sections it would require separate treatment, which we do not attempt here.

Definition 8.3 (Comparison maps at the boundary)**.** The algebras $\mathfrak{A}_{c_1 \neq 0}$ and $\mathfrak{A}_0 = \mathfrak{A}_{\mathrm{pre}}/\mathfrak{I}_{\mathrm{Weyl}}$ are not equal, so "$\omega_{c_1} \to \omega_0$" requires comparison maps. Let $\mathcal{O}_{\mathrm{reg}}$ be the *curvature-generated algebra*: the algebra generated by the smeared linearized Weyl tensor. For every c_1 (including 0) there is a canonical map $\iota_{c_1} : \mathcal{O}_{\mathrm{reg}} \to \mathfrak{A}_{c_1}$, because the Weyl observable annihilates both diffeomorphism gauge (Theorem 7.1(i)) and, at $c_1 = 0$, the emergent Weyl-null direction (ι_0 factors through the quotient by $\mathfrak{I}_{\mathrm{Weyl}}$). The boundary statement of Theorem 9.1 is

$$\omega_{c_1} \circ \iota_{c_1} \longrightarrow \omega_0 \circ \iota_0 \qquad \text{distributionally on } \mathcal{O}_{\mathrm{reg}}, \tag{13}$$

and *not* convergence on the complete split-phase observable algebra.

Remark 8.4 (Regularity holds on the quotient, not before it). The raw scalar kernel $W_S^+ = e^{-i\Omega t}/(2\mu_S\Omega)$ with $\mu_S = 3c_1/2$ diverges like $1/c_1$: the full reduced-coordinate kernel of the split phase has *no* regular limit. What converges is the family of presymplectic phase spaces

$$(\Gamma_{c_1}, \Omega_{c_1}) \longrightarrow (\Gamma_0/\ker\Omega_0,\ \bar{\Omega}_0), \tag{14}$$

together with the pulled-back functionals (13) on the curvature-generated algebra: the divergent direction is exactly the one that enters $\ker\Omega_0$, is annihilated by ι-observables, and is quotiented by the emergent Weyl symmetry. The observable algebra changes rank at the boundary; this is a structural feature of the conformal locus, not an artifact.

	± 2	± 1	0
$c_1 \neq 0$	PU pair	massive ghost	massive ghost
$c_1 = 0$	$\Box^2$ Jordan	massless ghost	Weyl gauge/null

Theorem 8.5 (Termination of the positive real form)**.** *The uniform positive completion is built on the split normal-mode decomposition of the TT PU pairs, whose rapidity is*

$$r(k, M) = \log\frac{(\omega_1+\omega_2)^2}{\omega_1^2-\omega_2^2} = \log\frac{\left(\sqrt{k^2+M^2}+k\right)^2}{M^2} = \log\frac{4k^2}{M^2} + O\!\left(\frac{M^2}{k^2}\right). \tag{15}$$

As $M \to 0$ at every fixed $k > 0$, $r \to \infty$ and the symplectic normal-mode transformation degenerates exactly as $\mathrm{cond}(N) = e^r\,(1 + o(1)) \sim 4k^2/M^2$ *(condition number in the Euclidean operator* 2*-norm in the canonical basis); equivalently, $\|\mu(S_+)\| \to \infty$: the positive representative escapes to infinity in* $\mathrm{Sp}(4,\mathbb{C})/\mathrm{Sp}(2)$ *(Section 5). At the boundary the dynamical map is a Jordan block and, by the Jordan obstruction of [6], no positive-definite invariant form exists. Precisely: the split-phase positive metric has* no continuation as a nondegenerate positive-definite invariant Hilbert metric *through* $c_1 = 0$*. This leaves open singular or positive-semidefinite boundary forms, and the non-diagonalizable PT-symmetric Jordan realization of Bender and Mannheim [3, 4] remains available* as a distinct, singular completion of the boundary theory *— it is not a continuation of the split Hilbert metric, whose similarity transformation itself becomes singular in the equal-frequency limit [3]. Only the Krein real form continues as a nondegenerate completion of the same complex canonical variables; the spectral functional continues distributionally on the curvature-generated algebra (Definition 8.3). (Numerically,* $\mathrm{cond}(N) = 7.6, 47.5, 403, 4447$ *at* $M = 1, 0.3, 0.1, 0.03$*,* $k = 1$*, with* $\mathrm{cond}(N)/e^r \to 1$*; this serves as a regression check of the exact law, not as the proof.)*

9 The master theorem

Theorem 9.1 (Classification)**.** *Consider free scalar-free Einstein–Weyl gravity linearized about Minkowski space. Let $\mathfrak{A}_{\mathrm{inv}}$ be the algebra of linear gauge-invariant observables, equivalently the* *reduced field algebra, with the additional Weyl-null quotient imposed at* $c_1 = 0$ *(Definition 2.3). Within the class of translation-invariant, quasifree, mode-local constructions (Definition 3.1) compatible with the reduced symplectic form, the spectral condition, and the Poincaré action on the physical polarization spaces, the following hold.*

For $c_1 \neq 0$*, the complex spectral covariance is unique: the reduced complex dynamical algebra* $(\mathcal{V}_{\mathbb{C}}, \sigma, U)$ *determines a unique Poincaré-covariant spectral quasifree functional on* $\mathfrak{A}_{\rm inv}$*. Up to* U*-equivariant complex-symplectic automorphisms preserving that covariance — in particular, modulo symplectic stabilizers commuting with the dynamics — there are exactly two admissible real-form classes (Definition 3.1):*

1. *a positive pseudo-Hermitian completion in which the full massive spin-*2 *multiplet has uniformly rotated reality; and*

2. *a Krein completion in which the gravitational field retains its standard reality and the massive spin-*2 *one-particle irrep carries uniform negative sign relative to the positive massless graviton sector.*

No completion assigning different signatures or real structures to different helicities of the massive spin-2 *irrep is Poincaré covariant. The two admissible real forms induce the same complex bilinear covariance, and hence the same complex correlation functions on* $\mathfrak{A}_{\rm inv}$*; they differ only in involution and completion (Remark 3.2).*

As $c_1 \to 0$*, the functionals pulled back to the curvature-generated regular subalgebra converge distributionally,* $\omega_{c_1} \circ \iota_{c_1} \to \omega_0 \circ \iota_0$ *(Definition 8.3); no convergence is claimed on the complete split-phase observable algebra, whose rank changes at the boundary. The transverse-traceless sector becomes a pair of* $\Box^2$ *Jordan systems, the vector sector becomes two ordinary second-order massless ghost modes, and the scalar sector enters the kernel of the presymplectic form and is removed by the emergent Weyl gauge symmetry. The split-phase positive metric has no continuation as a nondegenerate positive-definite invariant metric through this boundary (any* PT*-symmetric Jordan realization there is a distinct, singular completion), whereas the Krein real form continues as a nondegenerate completion of the same complex canonical variables to the resulting Jordan theory.*

Proof. Category: Definition 3.1. Uniqueness of the complex covariance on $\mathfrak{A}_{\rm inv}$: Theorems 6.1, 6.2 (residues forced by the reduced symplectic form and covariance; the potential representative is fixed exactly modulo pure-gauge kernels by (11)). Two real forms and no hybrid: Theorem 3.3 (modewise), Lemma 4.1 (lift to the Poincaré group; two involution classes), Theorems 4.2–4.3. Identity of complex correlators: Theorems 6.3, 7.1, Remark 3.2. Stabilizer statement: orbit constancy [8] with the enlarged stabilizers of Section 2 and the canonical representative of Proposition 5.1. Boundary: Theorems 8.1, 8.5, Definition 8.3, and Remark 8.4. □

10 Discussion

Relation to the two programs. The theorem places the positive-PT quantization [2, 4] and the Krein quantization on the two sides of a mathematically controlled classification: they are the two covariant real forms of one reduced complex spectral theory, agreeing on every complex gauge-invariant correlation function, and distinguished by which member of the trilemma they sacrifice — standard gravitational reality (positive form) or positivity of the norm (Krein form). A terminological caution: Bateman and Turok [5] quantize the scalar perfect-square theory, not this gravitational field; the second class is therefore a *Bateman–Turok-type Krein real form* — the gravitational extension of their construction — rather than their construction itself. The conformal locus is one-sided in the precise sense of Theorem 8.5: only the Krein real form continues as a nondegenerate completion of the same complex canonical variables, while the split-phase positive metric has no nondegenerate invariant positive-definite limit, and any PT-symmetric Jordan realization at the boundary [3] is a distinct, singular completion rather than a continuation of the split Hilbert metric. In this precise sense Mannheim's construction is a split-phase real form and the Bateman–Turok-type Krein form is its resonant continuation. One overreading must be excluded: agreement of the two real forms on every complex gauge-invariant correlation function is a *free-theory* statement, and does not make the completions interchangeable descriptions of an interacting theory. For the fourth-order scalar the companion interaction paper [9] proves they separate: the canonical positive metric is obstructed by on-shell branch-changing conversion while the Krein/ghost-parity structure survives the same interaction exactly. The corresponding gravitational computation has not been performed (see the outlook below).

Relation to Mannheim's CPT program. Mannheim has argued [10] that time-independent conservation of scalar products in a non-Hermitian relativistic theory forces spectra that are real or occur in complex-conjugate pairs together with an antilinear symmetry, which complex Lorentz invariance identifies with CPT; the underlying equivalence of pseudo-Hermiticity, antilinear symmetry, and real-or-conjugate-pair spectra (with matched Jordan data) is Mostafazadeh's [11]. None of that organization is claimed here; the contribution of this paper is the gauge-reduced helicity structure and its consequences. Placed against that program, the present classification says the following. (i) Away from the conformal locus every massive spin-2 polarization admits a positive completion, but only *jointly*: the trilemma and the no-hybrid theorem force one uniform choice across the multiplet, and the positive choice carries uniformly rotated massive reality (Theorems 3.3–4.3). A CPT-based positive scalar product in the sense of [10] therefore lies *inside* the classified positive real-form class — with the rotated-reality qualification made explicit here. (ii) At the conformal locus the positive representative escapes to infinity ($\mathrm{cond}(N) = e^r(1 + o(1))$, Theorem 8.5) and the TT dynamics becomes a Jordan block. (iii) What the Jordan theorem excludes there is exactly the class of nondegenerate, time-independent, positive-definite invariant Hermitian forms making the Jordan generator self-adjoint — and nothing more. The equal-frequency treatment of Bender and Mannheim [3] uses nonstationary Jordan states and a degenerate pairing, i.e. a construction *outside* the excluded class: there is no contradiction between the no-go and that realization, only a sharp delimitation. (iv) The trilemma is then the precise form of the choice a CPT-based ghost-free quadratic gravity must make away from the boundary: rotated massive reality with a positive norm, or standard gravitational reality with a Krein norm — no covariant hybrid exists.

The geometric picture. The Cartan formulation of Section 5 turns the phase diagram into geometry: the split phase is the locus where the positive representative sits at finite symmetric-space displacement $\|\mu(S_+)\| = \sqrt{2}\,r$ from the origin, the approach to the conformal locus is escape to infinity along the C_2 chamber wall, and the Jordan boundary is the ideal boundary point at which no positive-definite invariant form remains. Minimum distortion, orbit constancy, and the Jordan obstruction — imported here from the companions — are the value, the fibers, and the boundary behaviour of the single map μ restricted to the split orbit.

Boundary truncation is a third, distinct construction. Maldacena's boundary-condition selection of the Einstein branch from conformal gravity [13] removes one branch of the solution algebra, $\ker(L_1 L_2) \to \ker L_1$; the completions classified here retain the full fourth-order solution algebra and change its inner-product structure. The two mechanisms answer the ghost problem differently and should not be identified; on Einstein backgrounds with $\Lambda \neq 0$ the critical locus shifts [12] and pure conformal gravity contains a partially massless sector [14], so the flat-space phase diagram above is the $\Lambda \to 0$ slice of a richer picture we do not analyze here.

Interaction diagnostic (outlook). The free classification fixes the arena for the interacting question: for the positive real form, whether the rotated massive reality is compatible with the cubic Weyl vertices; for the Krein form, whether a gravitational ghost parity commutes with the interaction and with BRST symmetry, as its scalar analogue does in the perfect-square theory [5]. Both are sharp, computable questions — the natural first test is $[\mathcal{P}_{\mathrm{ghost}}, S_{\mathrm{int}}] = 0$ at cubic order — and they provide the first necessary tests of whether either real form supports an interacting unitarity statement; they do not by themselves settle loop anomalies, BRST cohomology, renormalization of the parity, or positivity of physical probabilities. The companion interaction paper [9] supplies an interaction-obstruction mechanism — on-shell branch-changing conversion obstructing the deformation of the positive metric — that can now be applied to the cubic Einstein–Weyl vertices; no interacting gravitational conclusion is assumed here. We have not performed these computations; the free theorem is neutral on the outcome.

References

[1] K. S. Stelle, Phys. Rev. D **16** (1977) 953.

[2] C. M. Bender and P. D. Mannheim, *No-ghost theorem for the fourth-order derivative Pais–Uhlenbeck oscillator model*, Phys. Rev. Lett. **100** (2008) 110402.

[3] C. M. Bender and P. D. Mannheim, *Exactly solvable PT-symmetric Hamiltonian having no Hermitian counterpart*, Phys. Rev. D **78** (2008) 025022, arXiv:0804.4190.

[4] P. D. Mannheim, *Solution to the ghost problem in higher-derivative gravity*, arXiv:2109.12743.

[5] S. Bateman and N. Turok, *Escape from Ostrogradsky via hidden ghost parity*, arXiv:2607.00096.

[6] *Canonical positive symplectic diagonalization of the Pais–Uhlenbeck oscillator*, companion paper and verification repository, 2026.

[7] *The Pais–Uhlenbeck metric as a minimum-distortion principle, and the representation problem for the fourth-order field*, companion paper, 2026.

[8] *The universal vacuum of the fourth-order scalar field: metric orbits, Fock sectors, and the Krein boundary*, companion paper, 2026.

[9] *Interaction obstructions, resonant PT breaking, and doubled Jordan symmetry in fourth-order theories*, companion paper (Paper V), 2026.

[10] P. D. Mannheim, *CPT symmetry without Hermiticity*, arXiv:1611.02100.

[11] A. Mostafazadeh, *Pseudo-Hermiticity for a class of nondiagonalizable Hamiltonians*, J. Math. Phys. **43** (2002) 6343, arXiv:math-ph/0207009.

[12] H. Lü and C. N. Pope, *Critical gravity in four dimensions*, Phys. Rev. Lett. **106** (2011) 181302, arXiv:1101.1971.

[13] J. Maldacena, *Einstein gravity from conformal gravity*, arXiv:1105.5632.

[14] S. Deser, E. Joung and A. Waldron, *Partial masslessness and conformal gravity*, J. Phys. A **46** (2013) 214019, arXiv:1208.1307.

[15] B. Holdom, Nucl. Phys. B **1000** (2024) 116696.

[16] R. J. Riegert, Phys. Lett. B **134** (1984) 56.

A Verification checks and the theorems they support

Scripts: `verify_gravity_reduction.py` (G1–G7), `verify_gravity_completion.py` (G8–G9), `verify_gravity_spectral.py` (G10–G12), each printing `ALL PASS`; shared engine `gravity_engine.py` (exact $O(\epsilon^2)$ second variation).

Check	Supports
G1–G2	TT perfect-square Lagrangian and PU block (Theorem 2.1)
G3	vector w-reduction, unpaired massive ghost (Theorem 2.2(i))
G4–G5	scalar reduction, scalaron decoupling at $\alpha = -3\beta$ (Theorem 2.2(ii))
G6	sector gauge invariance (Section 2)
G7	polarization-enhanced stabilizer $4 \to 16$ (Remark after Definition 2.3)
G8	quarter-turn identities, trilemma rows, coset $T_0 \cdot SO(2,\mathbb{C})$ (Theorem 3.3)
G9a–b	$SO(3)$ commutant $= \mathbb{R}I_5$, hybrid violation (Theorem 4.2)
G9c	$[J_i, D_{\text{tot}}] = 0$; quarter-turn is admissible positive diagonalizer (Theorem 4.3)
G10a–b	sector kernels, bisolution property, commutator normalizations (Theorem 6.1)
G10c	projector contractions $\frac{1}{2} : \frac{M^2}{2} : \frac{1}{6}$; absolute coefficient $C = 4/c_1$ (Theorem 6.2)
G10d	real-form independence of the complex kernel (Theorem 6.3)
G11a–c	Riemann kills pure gauge; Weyl traceless; $\Pi^{(2,M)}$–Π_0 contraction equality (Theorem 7.1(i)–(ii))
G11e	Weyl symbol nonzero on massless TT, kills null gauge (proof of Theorem 7.1(iv))
G12a, a′	TT Jordan limit incl. the $1/\gamma$ prefactor (Theorem 8.1(i))
G12b	vector massless limit, fixed μ_V (Theorem 8.1(ii))
G12c	Weyl shift is a symmetry iff $c_1 = 0$; scalar kinetic coefficient $\to 0$ (Theorem 8.1(iii))
G12d, d′	$\operatorname{cond}(N)$ divergence and exact law $\operatorname{cond}(N)/e^r \to 1$ (Theorem 8.5)

Checks establish identities; the representation-theoretic and microlocal statements (Lemma 4.1, existence of completions, WF equality) are analytic, as stated in the Verification paragraph.